Prompt 9:

Objective: Detect the incorporation of specific sustainable practices within the provided source code, focusing on green architectural tactics for machine learning. Analyze the code to identify direct and indirect sustainable approaches.

Task Description: 1) Review the source code to determine whether each of the following sustainable approaches mentioned or implemented:
1.Choose an energy efficient algorithm: Choose the most energy-efficient algorithm that achieves sufficient level of accuracy
2.Choose a lightweight algorithm alternative: If possible, choose lighter alternatives of existing algorithms
3.Decrease model complexity: Decrease the complexity of an ML model
4.Consider reinforcement learning for energy efficiency: Use reinforcement learning to optimize energy efficiency at run time
5.Use dynamic parameter adaptation: Design parameters that are dynamically adapted based on the input data
6.Use built-in library functions: Use built-in libraries for ML models if possible
7.Set energy consumption as a model constraint: Consider energy consumption as one predetermined parameter for optimizing the ML model
8.Consider graph substitution: Replace energy-intensive model parts with similar, but less energy-consuming parts
9.Enhance model sparsity: Reduce the number of model parameters or set their values to zero
10.Consider energy aware pruning: Prune neural networks starting from the most energy-intensive layer
11.Consider transfer learning: Use pre-trained ML models for other similar tasks
12.Consider knowledge distillation: Use knowledge from a large ML model to train a smaller model
13.Use quantization-aware training: Convert high-precision data types to lower precision during training
14.Use checkpoints during training: Use checkpoints to avoid a knowledge loss in case of a premature termination
15.Design for memory constraints: Consider possible memory constraints during training

2) Additionally, identify any new direct or indirect sustainable practices not listed above that are present in the code.
Additional Instructions:
Focus on specific lines or sections that relate to the listed sustainable practices.
Clearly state whether each approach is present, supported by specific code snippets.
If new approaches are not present, only write "No" without describing the code.

Reporting Instructions:
Structure the output as follows:

Presence of Approaches: For each approach found in the source code, specify its occurrence and provide the relevant code snippet. If not present, write “No”.
New Approaches: Analyze the code to identify new direct and indirect sustainable practices. If a new approach is found, write "Yes" including the code and a brief description. If not present, write “No”.

Examples:
"Design for memory constraints: No"
"Use built-in library function: import torch"
"New Approaches: No"
"New Approaches: Yes, Implementing Early Stopping early_stopping = EarlyStopping(monitor='val_loss')"

Prompt 8:

Objective: Detect the incorporation of specific sustainable practices within the provided source code, focusing on green architectural tactics that improve efficiency and minimize resource consumption.

Task Description: Review the source code to determine whether each of the following sustainable approaches is explicitly mentioned or implemented:

1. Choose an energy efficient algorithm: Choose the most energy-efficient algorithm that achieves sufficient level of accuracy
2. Choose a lightweight algorithm alternative: If possible, choose lighter alternatives of existing algorithms
3. Decrease model complexity: Decrease the complexity of an ML model
4. Consider reinforcement learning for energy efficiency: Use reinforcement learning to optimize energy efficiency at run time
5. Use dynamic parameter adaptation: Design parameters that are dynamically adapted based on the input data
6. Use built-in library functions: Use built-in libraries for ML models if possible
7. Set energy consumption as a model constraint: Consider energy consumption as one predetermined parameter for optimizing the ML model
8. Consider graph substitution: Replace energy-intensive model parts with similar, but less energy-consuming parts
9. Enhance model sparsity: Reduce the number of model parameters or set their values to zero
10. Consider energy aware pruning: Prune neural networks starting from the most energy-intensive layer
11. Consider transfer learning: Use pre-trained ML models for other similar tasks
12. Consider knowledge distillation: Use knowledge from a large ML model to train a smaller model
13. Use quantization-aware training: Convert high-precision data types to lower precision during training
14. Use checkpoints during training: Use checkpoints to avoid a knowledge loss in case of a premature termination
15. Design for memory constraints: Consider possible memory constraints during training

Reporting Instructions:

Be concise and short to collect the information as data for a larger-scale analysis.

Structure the output as follows:

Presence of Approaches: Provide relevant code snippet. If not present, write “No”.

Additional Findings: Describe any newly identified sustainable practices and their contributions only if you find it.

No approaches: If no approaches and no additional findings, write “None”.

An example is "Design for memory constraints: No"

Prompt 7:

Objective: Detect the incorporation of specific sustainable practices within the provided source code, focusing on green architectural tactics that improve efficiency and minimize resource consumption.

Task Description: Below is a list of sustainable approaches, each accompanied by a detailed description. Review the source code provided to determine whether each approach is explicitly mentioned or implemented.

16. Choose an energy efficient algorithm: Choose the most energy-efficient algorithm that achieves sufficient level of accuracy
17. Choose a lightweight algorithm alternative: If possible, choose lighter alternatives of existing algorithms
18. Decrease model complexity: Decrease the complexity of an ML model
19. Consider reinforcement learning for energy efficiency: Use reinforcement learning to optimize energy efficiency at run time
20. Use dynamic parameter adaptation: Design parameters that are dynamically adapted based on the input data
21. Use built-in library functions: Use built-in libraries for ML models if possible
22. Set energy consumption as a model constraint: Consider energy consumption as one predetermined parameter for optimizing the ML model
23. Consider graph substitution: Replace energy-intensive model parts with similar, but less energy-consuming parts
24. Enhance model sparsity: Reduce the number of model parameters or set their values to zero
25. Consider energy aware pruning: Prune neural networks starting from the most energy-intensive layer
26. Consider transfer learning: Use pre-trained ML models for other similar tasks
27. Consider knowledge distillation: Use knowledge from a large ML model to train a smaller model
28. Use quantization-aware training: Convert high-precision data types to lower precision during training
29. Use checkpoints during training: Use checkpoints to avoid a knowledge loss in case of a premature termination
30. Design for memory constraints: Consider possible memory constraints during training

Reporting Instructions:

- Be concise and short to collect the information as data for a larger-scale analysis.
- Structure the output as follows:

Presence of Approaches: For each approach found in the source code, specify its occurrence and provide the relevant code snippet.

Non-Existent Approaches: If an approach is not present, only write “No”.

Additional Findings: If other sustainable or green architectural tactics not listed here are identified, describe them and explain how they contribute to efficiency and reduced resource usage.

No approaches: If none of the approaches are present and no additional approaches are found, only write “None”.

Prompt 6:

Objective: The objective of this analysis is to detect the incorporation of specific sustainable practices within the provided source code. This assessment focuses on green architectural tactics that improve efficiency and minimize resource consumption.
Task Description: Below is a list of sustainable approaches, each accompanied by a detailed description. Review the source code provided to determine whether each approach is explicitly mentioned or implemented.

1. Choose an energy efficient algorithm: Choose the most energy-efficient algorithm that achieves sufficient level of accuracy
2. Choose a lightweight algorithm alternative: If possible, choose lighter alternatives of existing algorithms
3. Decrease model complexity: Decrease the complexity of an ML model
4. Consider reinforcement learning for energy efficiency: Use reinforcement learning to optimize energy efficiency at run time
5. Use dynamic parameter adaptation: Design parameters that are dynamically adapted based on the input data
6. Use built-in library functions: Use built-in libraries for ML models if possible
7. Set energy consumption as a model constraint: Consider energy consumption as one predetermined parameter for optimizing the ML model
8. Consider graph substitution: Replace energy-intensive model parts with similar, but less energy-consuming parts
9. Enhance model sparsity: Reduce the number of model parameters or set their values to zero
10. Consider energy aware pruning: Prune neural networks starting from the most energy-intensive layer
11. Consider transfer learning: Use pre-trained ML models for other similar tasks
12. Consider knowledge distillation: Use knowledge from a large ML model to train a smaller model
13. Use quantization-aware training: Convert high-precision data types to lower precision during training
14. Use checkpoints during training: Use checkpoints to avoid a knowledge loss in case of a premature termination
15. Design for memory constraints: Consider possible memory constraints during training

Reporting Instructions:
Please be concise and short to collect the information as data for a larger scale analysis.

The output should be structured in the following ways:

Presence of Approaches: For each approach that is found in the source code, specify its occurrence and provide the relevant code snippet.
Non-Existent Approaches: If an approach is not present, only say “No”.
Additional Findings: If other sustainable or green architectural tactics not listed here are identified, describe them and explain how they contribute to efficiency and reduced resource usage.
No approaches: If there are not presented the approaches and you don’t find additional approaches only say “None”

Prompt 5:

Objective: The objective of this analysis is to detect the incorporation of specific sustainable practices within the provided source code. This assessment focuses on green architectural tactics that improve efficiency and minimize resource consumption.
Task Description: Below is a list of sustainable approaches, each accompanied by a detailed description. Review the source code provided to determine whether each approach is explicitly mentioned or implemented.
Choose an energy efficient algorithm: Choose the most energy-efficient algorithm that achieves sufficient level of accuracy
Choose a lightweight algorithm alternative: If possible, choose lighter alternatives of existing algorithms
Decrease model complexity: Decrease the complexity of an ML model
Consider reinforcement learning for energy efficiency: Use reinforcement learning to optimize energy efficiency at run time
Use dynamic parameter adaptation: Design parameters that are dynamically adapted based on the input data
Use built-in library functions: Use built-in libraries for ML models if possible
Set energy consumption as a model constraint: Consider energy consumption as one predetermined parameter for optimizing the ML model
Consider graph substitution: Replace energy-intensive model parts with similar, but less energy-consuming parts
Enhance model sparsity: Reduce the number of model parameters or set their values to zero
Consider energy aware pruning: Prune neural networks starting from the most energy-intensive layer
Consider transfer learning: Use pre-trained ML models for other similar tasks
Consider knowledge distillation: Use knowledge from a large ML model to train a smaller model
Use quantization-aware training: Convert high-precision data types to lower precision during training
Use checkpoints during training: Use checkpoints to avoid a knowledge loss in case of a premature termination
Design for memory constraints: Consider possible memory constraints during training

Reporting Instructions:
Presence of Approaches: For each approach that is found in the source code, specify its occurrence and provide the relevant code snippet.
Non-Existent Approaches: If an approach is not present, omit it from the report.
Additional Findings: If other sustainable or green architectural tactics not listed here are identified, describe them and explain how they contribute to efficiency and reduced resource usage.
No approaches: If there are not presented the approaches and you don't find additional approaches only say "None"

Prompt 4:

Purpose: This analysis aims to identify the implementation of specific sustainable approaches in the source code of a project, focusing on green architectural tactics to enhance efficiency and reduce resource usage.
Task: Below is a list of sustainable approaches along with detailed descriptions. Please examine the source code and determine whether each approach is mentioned or implemented.
Choose an energy efficient algorithm: Choose the most energy-efficient algorithm that achieves sufficient level of accuracy
Choose a lightweight algorithm alternative: If possible, choose lighter alternatives of existing algorithms
Decrease model complexity: Decrease the complexity of an ML model
Consider reinforcement learning for energy efficiency: Use reinforcement learning to optimize energy efficiency at run time
Use dynamic parameter adaptation: Design parameters that are dynamically adapted based on the input data
Use built-in library functions: Use built-in libraries for ML models if possible
Set energy consumption as a model constraint: Consider energy consumption as one predetermined parameter for optimizing the ML model
Consider graph substitution: Replace energy-intensive model parts with similar, but less energy-consuming parts
Enhance model sparsity: Reduce the number of model parameters or set their values to zero
Consider energy aware pruning: Prune neural networks starting from the most energy-intensive layer
Consider transfer learning: Use pre-trained ML models for other similar tasks
Consider knowledge distillation: Use knowledge from a large ML model to train a smaller model
Use quantization-aware training: Convert high-precision data types to lower precision during training
Use checkpoints during training: Use checkpoints to avoid a knowledge loss in case of a premature termination
Design for memory constraints: Consider possible memory constraints during training

Instructions:
For each approach, indicate clearly whether it is 'Present' or 'Not Found'.
If an approach is found, can you specify where it appears and write the code used.

If an approach is not present, simply avoid to write.
If you find other sustainable or green approaches and green architectural tactics to improve efficiency and reduce resource use that were not mentioned, you can say what they are and give an explanation.

Prompt 3:

Purpose: This analysis aims to identify the implementation of specific sustainable approaches in the source code of a project, focusing on green architectural tactics to enhance efficiency and reduce resource usage.
Task: Below is a list of sustainable approaches along with detailed descriptions. Please examine the source code and determine whether each approach is mentioned or implemented.
Choose an energy efficient algorithm: Choose the most energy-efficient algorithm that achieves sufficient level of accuracy
Choose a lightweight algorithm alternative: If possible, choose lighter alternatives of existing algorithms
Decrease model complexity: Decrease the complexity of an ML model
Consider reinforcement learning for energy efficiency: Use reinforcement learning to optimize energy efficiency at run time
Use dynamic parameter adaptation: Design parameters that are dynamically adapted based on the input data
Use built-in library functions: Use built-in libraries for ML models if possible
Set energy consumption as a model constraint: Consider energy consumption as one predetermined parameter for optimizing the ML model
Consider graph substitution: Replace energy-intensive model parts with similar, but less energy-consuming parts
Enhance model sparsity: Reduce the number of model parameters or set their values to zero
Consider energy aware pruning: Prune neural networks starting from the most energy-intensive layer
Consider transfer learning: Use pre-trained ML models for other similar tasks
Consider knowledge distillation: Use knowledge from a large ML model to train a smaller model
Use quantization-aware training: Convert high-precision data types to lower precision during training
Use checkpoints during training: Use checkpoints to avoid a knowledge loss in case of a premature termination
Design for memory constraints: Consider possible memory constraints during training

Instructions:
For each approach, indicate clearly whether it is 'Present' or 'Not Found'.
If an approach is found, can you specify where it appears and write the code used.
If an approach is not present, simply state "Not Found".
If you find other sustainable or green approaches and green architectural tactics to improve efficiency and reduce resource use that were not mentioned, you can say what they are and give an explanation.

Prompt 2:

Purpose: This analysis aims to identify the implementation of specific sustainable approaches in the source code of a project, focusing on green architectural tactics to enhance efficiency and reduce resource usage.
Task: Below is a list of sustainable approaches along with detailed descriptions. Please examine the source code and determine whether each approach is mentioned or implemented.
Choose an energy efficient algorithm: Choose the most energy-efficient algorithm that achieves sufficient level of accuracy
Choose a lightweight algorithm alternative: If possible, choose lighter alternatives of existing algorithms
Decrease model complexity: Decrease the complexity of an ML model
Consider reinforcement learning for energy efficiency: Use reinforcement learning to optimize energy efficiency at run time
Use dynamic parameter adaptation: Design parameters that are dynamically adapted based on the input data
Use built-in library functions: Use built-in libraries for ML models if possible
Set energy consumption as a model constraint: Consider energy consumption as one predetermined parameter for optimizing the ML model
Consider graph substitution: Replace energy-intensive model parts with similar, but less energy-consuming parts
Enhance model sparsity: Reduce the number of model parameters or set their values to zero
Consider energy aware pruning: Prune neural networks starting from the most energy-intensive layer
Consider transfer learning: Use pre-trained ML models for other similar tasks
Consider knowledge distillation: Use knowledge from a large ML model to train a smaller model
Use quantization-aware training: Convert high-precision data types to lower precision during training
Use checkpoints during training: Use checkpoints to avoid a knowledge loss in case of a premature termination
Design for memory constraints: Consider possible memory constraints during training

Instructions:
For each approach, indicate clearly whether it is 'Present' or 'Not Found'.
If an approach is found, can you specific where it appears and write the code used.
If an approach is not present, simply state "Not Found".
If you find other sustainable or green approaches and green architectural tactics to improve efficiency and reduce resource use that were not mentioned, you can say what they are and give an explanation.

Prompt 1:

I have a file containing the source code of a project, and I need to determine if specific sustainable approaches related to green architectural tactics are mentioned or implemented. Below is the list of approaches and their description that need to be checked:

Choose an energy efficient algorithm: Choose the most energy-efficient algorithm that achieves sufficient level of accuracy
Choose a lightweight algorithm alternative: If possible, choose lighter alternatives of existing algorithms
Decrease model complexity: Decrease the complexity of an ML model
Consider reinforcement learning for energy efficiency: Use reinforcement learning to optimize energy efficiency at run time
Use dynamic parameter adaptation: Design parameters that are dynamically adapted based on the input data
Use built-in library functions: Use built-in libraries for ML models if possible
Set energy consumption as a model constraint: Consider energy consumption as one predetermined parameter for optimizing the ML model
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Consider knowledge distillation: Use knowledge from a large ML model to train a smaller model
Use quantization-aware training: Convert high-precision data types to lower precision during training
Use checkpoints during training: Use checkpoints to avoid a knowledge loss in case of a premature termination
Design for memory constraints: Consider possible memory constraints during training

Please analyze the text and report whether each approach is in the file. Indicate clearly for each approach whether it is 'Present' or 'Not Found'. Provide a brief note on the context or usage if an approach is found. If the approach is not present, indicate only "Not Found."